

4.1 Basic Modular Arithmetic - 1

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Modular arithmetic means arithmetic using the "modulo" operator. Adjusts division to only care about the remainder.

Definition: If n and m are integers with $m \neq 0$, we say m divides n iff there is an integer q such that $n = m \cdot q$ or $\frac{n}{m} = q$. This is denoted $m | n$. **Order matters!** $m | n \neq n | m$.
denom. | numer.

Definition: If n and m are integers and $m | n$, then m is a factor or divisor of n , and n is a multiple of m .

Does $3 | 7$? $7 | 3$? $4 | 8$? $15 | 3$?
 $\frac{7}{3} \notin \mathbb{Z}$ $\frac{3}{7} \notin \mathbb{Z}$ $\frac{8}{4} = 2$ $\frac{3}{15} \notin \mathbb{Z}$
No. No. Yes. No.

If $n, d \in \mathbb{Z}^+$, how many $k \in \mathbb{Z}^+$ are there such that $d | k$ and $k \leq n$?

Since $k \in \mathbb{Z}^+$ and $k \leq n$, $1 \leq k \leq n$.

Since $d | k$ there is a $q \in \mathbb{Z}$ such that

$k = dq$. Then there are as many k

as there are q .

$$1 \leq k \leq n \Leftrightarrow 1 \leq dq \leq n \Leftrightarrow \frac{1}{d} \leq q \leq \frac{n}{d}.$$

Since q is an integer $\frac{1}{d} \leq q \Leftrightarrow \lceil \frac{1}{d} \rceil \leq q$

$$\Leftrightarrow 1 \leq q. \text{ Similarly } q \leq \frac{n}{d} \Leftrightarrow q \leq \lfloor \frac{n}{d} \rfloor$$

so $q \in \{1, 2, 3, \dots, \lfloor \frac{n}{d} \rfloor\}$. Then there are

$\lfloor \frac{n}{d} \rfloor$ many q thus $\lfloor \frac{n}{d} \rfloor$ many k .

Theorem: Let $a, b, c \in \mathbb{Z}$, $a \neq 0$. Then

1) If $a | b$ and $a | c$, then $a | (b+c)$

2) If $a | b$, then $a | bc$ for any c .

2) If $a|b$, then $a|bc$ for any c .

3) If $a|b$ and $b|c$, then $a|c$.

3) Let $a, b, c \in \mathbb{Z}$, $a \neq 0$ and suppose $a|b$ and $b|c$. Then there are integers x, y such that $b = a \cdot x$ and $c = b \cdot y$.

Then $c = b \cdot y = (a \cdot x) \cdot y = a \cdot (x \cdot y)$

so c is a multiple of a (or $a|c$).

Therefore if $a|b$ and $b|c$, then $a|c$ ■

Corollary: If $a, b, c \in \mathbb{Z}$, $a \neq 0$, where $a|b$ and $b|c$, then for any integers x, y , $a|(bx + cy)$.

Proof Outline: $a|b$ and $b|c \Rightarrow \exists k, l \in \mathbb{Z}$

so that $b = a \cdot k$ and $c = b \cdot l$. Let $x, y \in \mathbb{Z}$,

$$\begin{aligned} \text{then } b \cdot x + c \cdot y &= (ak)x + (bl)y \\ &= a(kx) + ((a \cdot k) l)y \\ &= a(kx) + a(k \cdot l \cdot y) \\ &= a(kx + kly) \end{aligned}$$

So $a|bx + cy$.

Theorem (The Division Algorithm):

Let $a, d \in \mathbb{Z}$, $d > 0$. Then there are unique

integers r, q such that $a = d \cdot q + r$

where $0 \leq r < d$.

This is not really an algorithm, but we keep its original name. It does not describe any steps or process, only that an integer may be expressed $a = dq + r$.

Definition: If $a, d, q, r \in \mathbb{Z}$ with

$d > 0$ and $0 \leq r < d$, and

$a = dq + r$, then d is the divisor

q is the quotient r is the

$a = dq + r$, then d is the divisor
 q is the quotient r is the
remainder, and a is the dividend.

Def: The modulo (or mod) operator
 is written $a \bmod d$ or $a \% d$ and gives
 the remainder when dividing a by d .

Def: If $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$ then
 a is congruent to b modulo m if $m \mid a - b$, or
 $(a \bmod m) = (b \bmod m)$. We write this
 as $a \equiv b \pmod{m}$. In this case, m
 is the modulus.

What are the quotient and remainder
 when 789 is divided by 21?

$$\begin{array}{r} 37 \\ 21 \overline{) 789} \\ \underline{- 63} \\ 159 \\ \underline{- 147} \\ 12 \end{array}$$

quotient is 37
 remainder is 12

What is $789 \bmod 21$?

$$789 \bmod 21 = 12$$

Find 3 integers congruent to $789 \pmod{21}$

Same remainder $\Rightarrow$ add a multiple of 21
 to 12

$$12 \equiv 789 \pmod{21} \quad \{ 19 + 21 = 40 \}$$

$$33 \equiv 789 \pmod{21} \quad \{ 40 + 21 = 61 \}$$

$$54 \equiv 789 \pmod{21} \quad \{ 19 - 21 = -2 \}$$

$$-9 \equiv 789 \pmod{21}$$

If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$,
 is it true that $a \equiv c \pmod{m}$?

$$a \equiv b \pmod{m} \Rightarrow m \mid a - b, \quad b \equiv c \Rightarrow m \mid b - c$$

Since $m \mid a - b$ and $m \mid b - c$, then

$$m \mid (a-b) + (b-c) \Rightarrow m \mid a-c$$

$$\Rightarrow a \equiv c \pmod{m}.$$

Theorem: For $m \in \mathbb{Z}^+$, $a, b \in \mathbb{Z}$, then

$a \equiv b \pmod{m}$ iff $a = b + km$ for some integer k .

Theorem: Let $m \in \mathbb{Z}^+$, $a, b, c, d \in \mathbb{Z}$.

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$,

then $a+c \equiv b+d \pmod{m}$ and

$$a \cdot c \equiv b \cdot d \pmod{m}.$$

Verify the congruence $2 \equiv 14 \pmod{12}$.

Is it true that $1 \equiv 7 \pmod{12}$? No.

So we must be careful when dividing congruences.

$$-1 \equiv 2 \pmod{3} \text{ and } 1 \equiv 4 \pmod{3}$$

$$\text{is } (-1)' \equiv 2^4 \pmod{3} \text{? No.}$$

$$(-1)' = -1 = -1 \cdot 3 + 2 \text{ so } -1 \pmod{3} = 2.$$

$$(2)^4 = 16 = 5 \cdot 3 + 1 \text{ so } 2^4 \pmod{3} = 1$$

Corollary: Let $m \in \mathbb{Z}^+$ and $a, b \in \mathbb{Z}$, then

- $(a+b) \pmod{m} = ((a \pmod{m}) + (b \pmod{m})) \pmod{m}$
- $(a \cdot b) \pmod{m} = ((a \pmod{m})(b \pmod{m})) \pmod{m}.$

$$\text{Find } (9^5 + 7 \cdot 81) \pmod{4}$$

$$= (9^5 \pmod{4} + (7 \cdot 81) \pmod{4}) \pmod{4}$$

$$= ((9 \pmod{4})^5 \pmod{4} + (7 \pmod{4})(81 \pmod{4}) \pmod{4}) \pmod{4}$$

$$= (1^5 \pmod{4} + (3) \cdot 1 \pmod{4}) \pmod{4}$$

$$= (1+3) \pmod{4} = 0$$

Def: For $a \in \mathbb{Z}$, $m \in \mathbb{Z}^+$ if there is an integer x such that

$$x \cdot a \equiv 1 \pmod{m}$$

then $x = a^{-1}$ and x is called the

multiplicative inverse of a mod m.

$$2^{-1} \bmod 3$$

⇒ Want x such that

$$x \cdot 2 \equiv 1 \pmod{3}.$$

Try $x=2$. Then $(2)(2)=4 \equiv 1 \pmod{3}$
so $(2)^{-1} \equiv 2 \pmod{3}$

Inverses might not exist!

$2^{-1} \bmod 12$ is an integer x

such that $x \cdot 2 \equiv 1 \pmod{12}$.

In order for $x \cdot 2 \equiv 1 \pmod{12}$, need

$x \cdot 2$ to be odd but this is impossible

so $2^{-1} \bmod 12$ DOES NOT EXIST.

Corollary: Let $a \in \mathbb{Z}$, $m \in \mathbb{Z}^+$

then for any $N \in \mathbb{N}$

$$a^N \bmod m = (a^{-1} \bmod m)^N \bmod m$$

Evaluate $7^{-3} \bmod 4$.

$$7^{-3} \bmod 4 = (7^{-1} \bmod 4)^3 \bmod 4$$

where $7^{-1} = x$ with

$$x \cdot 7 \equiv 1 \pmod{4}. \text{ (Try } x=3!)$$

$$3 \cdot 7 = 21 = 5(4) + 1 \equiv 1 \pmod{4}$$

Thus $7^{-1} \equiv 3 \pmod{4}$.

$$\text{Then } (7^{-1} \bmod 4)^3 \bmod 4$$

$$= (3 \bmod 4)^3 \bmod 4$$

$$= 3^3 \bmod 4 = 27 \bmod 4 = \boxed{3}$$